Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 5: The Inductor and a summary of the LTI elements

M. Mert Ankaralı

Colour code used in these notes and in the slides

flux linkage φ | current i | voltage v | charge q | power p, energy w

1. The other way to store energy

Lecture 4 gave us an element that dissipates nothing: the capacitor holds energy in the electric field between its plates and hands it back when asked. Today's element does the same job with the other field — drive a current through a coil of wire and the magnetic field that appears holds energy too.

Lecture 4: capacitor
charge on plates, *electric* field
$i = C\,dv/dt$

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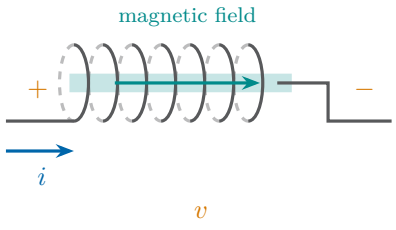
Today: inductor
current in a coil, *magnetic* field
$v = L\,di/dt$

The two terminal equations above are the same equation with v and i exchanged. That is no coincidence: almost everything proved about the capacitor becomes true of the inductor once you swap voltage for current. We will still build it from the ground up — but try to guess each result before it appears.

2. The coil, and the voltage it pushes back with

Start from something you can watch on a bench without knowing any field theory. Wind a length of wire into a coil and drive a current through it.

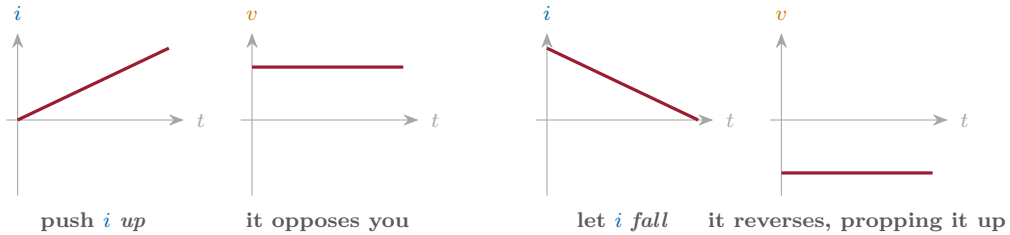
Two words: the **coil** is the object you just wound; the **inductor** is the circuit element we model it with, and that is the word we use from here on. (A real coil is an inductor plus the resistance of its own wire.)



A current in the coil sets up a magnetic field running through it. Nothing surprising yet — that happens around any wire.

The interesting part is what happens when you try to *change* that current.

Push the current up and the inductor develops a voltage that *opposes* you; let it fall and that voltage reverses, now trying to keep the current going. The faster the change, the larger the voltage.



Back EMF

The voltage an inductor develops when its current changes is called the **back EMF** ("electromotive force" is a historical name — it is a voltage). It always acts so as to *oppose the change* that produced it: it fights a rising current and props up a falling one.

Keep the **passive sign convention** and the marks never move: with i referenced into the + terminal, a rising current simply makes v positive and a falling one makes it negative. The opposition shows up in the *sign* of v, never in a redrawn polarity.

3. Inductance: how hard it pushes back

Two inductors in the same circuit will not push back equally. The constant of proportionality in $v \propto di/dt$ is the **inductance** L:

$$\boxed{v = L\frac{di}{dt}} \qquad 1\,\mathrm{H} = 1\,\frac{\mathrm{V}\cdot\mathrm{s}}{\mathrm{A}}.$$

Read the unit as the definition: one henry is the inductance that produces one volt of back EMF when the current is rising at one ampere per second.

L grows with

- the **number of turns** — each extra turn catches the field again, so L grows faster than the turn count itself;
- the **shape**: long and thin, or short and fat, differs even for the same wire;
- and above all **what is inside it** — an iron core instead of air multiplies L by hundreds.

One henry is large

A 1000-turn coil, $1\,\mathrm{cm}^2$ in section and 5 cm long, wound on air, comes to about 2.5 mH. Practical values run from nH to a few mH; reaching henries means an iron core. Compare Lecture 4: one farad was enormous for exactly the same reason.

4. Volt-seconds: the inductor's version of charge

We have the terminal equation, but not yet the *characteristic* — the curve that says what kind of element this is. Lecture 4 found the capacitor's by asking what was being accumulated: $q = \int i\,d\tau$, in *amp-seconds*, which we call coulombs. The inductor asks the same of the other variable. Integrate the voltage:

Flux linkage

$$\varphi(t) = \varphi(t_0) + \int_{t_0}^{t} v(\tau)\,d\tau \qquad \Longleftrightarrow \qquad v = \frac{d\varphi}{dt}$$

That integral is measured in *volt-seconds*, and a volt-second has a name of its own: the **weber** (Wb).

	capacitor (Lecture 4)	**inductor** (today)
accumulate which variable?	the current i	the voltage v
the running total	$q = \int i\,d\tau$	$\varphi = \int v\,d\tau$
measured in	amp-seconds = coulombs	volt-seconds = webers
so	$i = dq/dt$	$v = d\varphi/dt$

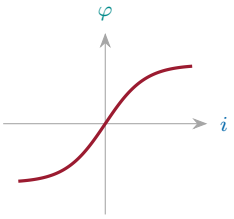
4.1. Why it is called *flux linkage*

Because φ is the magnetic field inside the winding, counted once for every turn it passes through. More turns, a fatter winding, or iron instead of air all mean more field and so a larger φ — the same three things that made L large. And a *changing* φ is the back EMF, which is Faraday's law.

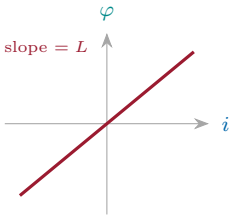
One difference from Lecture 4 is worth naming. Charge is countable stuff — so many coulombs of electrons on a plate. Volt-seconds are not a substance and nothing is piling up anywhere; φ is a bookkeeping variable. But it behaves, in every equation from here on, exactly as charge did.

Inductor

A two-terminal element is an **inductor** if, at every instant, its (i, φ) pairs lie on a curve in the i–φ plane.



an inductor in general



the LTI inductor

The bent curve is a real iron-cored inductor: past a certain current the core **saturates** and no extra flux can be squeezed into it. In this course every inductor is **linear and time-invariant**, so from here on the characteristic is the straight line.

And again, note what has *not* happened. There is no relation between the values of v and i at one instant: no function f or g with

$$v(t) = f\big(i(t)\big) \qquad \text{or} \qquad i(t) = g\big(v(t)\big).$$

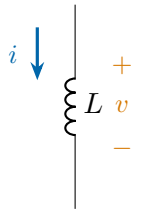
Fix the current and the voltage is still free — it depends on how i is *changing*, not on what it is. An inductor is not a resistor either, and it has no characteristic in the v–i plane.

5. The LTI inductor

LTI inductor

The i–φ characteristic is a fixed straight line through the origin, $\varphi = L\,i$ — and its slope is the same L we met in Section 3. Differentiating and using $v = d\varphi/dt$ gives back the terminal equation, now with a characteristic behind it:

$$\boxed{v \;=\; L\,\frac{di}{dt}} \qquad \text{or, integrated,} \qquad \boxed{i(t) = i(t_0) + \frac{1}{L}\int_{t_0}^{t} v(\tau)\,d\tau}$$



Two equations, two readings. **Differential**: the voltage is whatever the *rate of change* of current demands. **Integral**: the current is the running total of everything the voltage has done since t_0.

Four things that follow — each one the capacitor's, mirrored

- **At DC it is a short circuit.** i constant $\Rightarrow di/dt = 0 \Rightarrow v = 0$, whatever the current is. To a DC source, an inductor is just a piece of wire.
- **Its current is continuous.** Flip a switch, step the source: the inductor's current is the *same* an instant after as an instant before. Changing it takes time, because a jump would need an infinite voltage.
- **It remembers.** $i(t)$ depends on the whole history of v, not on its present value.
- **It holds energy**, and the amount is $w = \frac{1}{2}Li^2$ — fixed by the present current alone. Section 6 shows why, and that all of it can be taken back out.

Example 5.1 — a constant voltage across an inductor

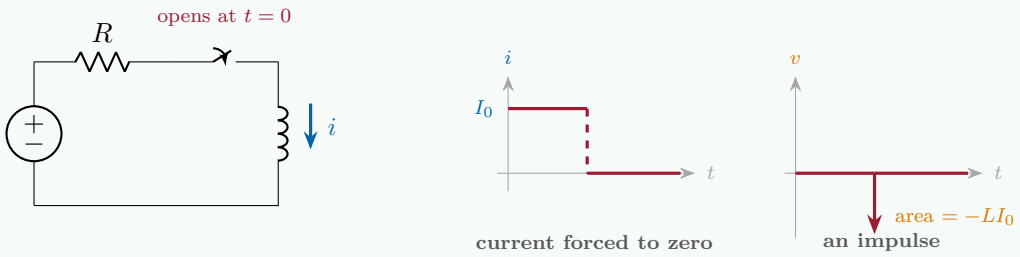
Hold $v = 5\,\text{V}$ across $L = 10\,\text{mH}$, starting from $i(0) = 0$. The integral form gives

$$i(t) = \frac{1}{L}\int_0^t 5\,d\tau = \frac{5}{10^{-2}}\,t = 500\,t \;\;[\text{A}].$$

A **ramp**: the current climbs at 500 A/s, reaching 5 A after 10 ms. Nothing about i can be read off v at one instant — only from the area under it. This is Example 4.1 with the variables swapped, and it is why a real inductor across a battery ends up limited by the resistance of its own wire.

Example 5.2 — opening a switch, and where the spark comes from

An inductor carries a steady current I_0. Now open a switch in series with it: the current is forced to 0 in zero time.



A step of $-I_0$ in the current means $di/dt = -I_0\,\delta(t)$, so

$$v(t) \;=\; L\,\frac{di}{dt} \;=\; -LI_0\,\delta(t),$$

an impulse of area $-LI_0$. This is the exact mirror of Example 4.2, where an ideal voltage step across a capacitor demanded a current impulse — and it is just as much an idealisation too far. What really happens is that the voltage across the opening contacts climbs until the air between them breaks down, and the stored $\frac{1}{2}LI_0^2$ leaves as a spark. The effect is a nuisance in switches and relays, and the whole point of an ignition coil.

6. Power and energy in an LTI inductor

Follow the current up from zero and the stored energy falls out on its own.

At $t = 0$ the current is zero. To get current flowing you have to raise it — and the moment you do, the inductor answers with its back EMF $v = L\,di/dt$, pointing *against* you. So whatever is driving the circuit is pushing current through an element that is pushing back, which means it is doing work, at a rate $p = v\,i = L\,i\,di/dt$. Add that up from $i = 0$ to $i = I$, changing the variable

of integration from t to i:

$$\int_0^t L\,i\,\frac{di}{d\tau}\,d\tau \;=\; \int_0^I L\,i\,di \qquad\Longrightarrow\qquad \boxed{w \;=\; \tfrac{1}{2}L\,i^{\,2} \;=\; \frac{\varphi^{\,2}}{2L}}$$

No resistance was involved, so none of that work turned into heat: it is still there, and the amount depends only on the present current. Differentiated, the same statement reads $p = \frac{d}{dt}(\frac{1}{2}Li^2)$ — power is the rate of change of the stored energy.

Now run it backwards. Let the current fall: $di/dt < 0$, so v changes sign, and the inductor stops resisting the current and starts *driving* it. p is negative, and the element hands back exactly what it was given. Example 5.2 is that handover done violently — all of it at once, into a spark.

$w \geq 0$ always, so an inductor can never give out more than it has been given: it is **passive**. And nothing is lost on the way in or on the way out: **lossless**.

So where was the energy kept?

In the **magnetic field** the current set up inside the winding — not in the wire, and not in the moving charge itself. That is the only difference from the capacitor, which put its work into the *electric* field between its plates. A field can hold energy without consuming any, and that is the whole reason these two elements are lossless while a resistor is not.

If you want the field version: a magnetic field of strength B carries $B^2/2\mu$ joules per cubic metre. Integrate that over the volume inside the winding and it comes to $\frac{1}{2}Li^2$ — the same answer, the long way round.

For the mechanically minded

Lecture 4 matched the capacitor to a moving mass. The inductor completes the picture:

$$v = L\,\frac{di}{dt} \qquad\longleftrightarrow\qquad u = \frac{1}{k}\,\frac{dF}{dt},$$

which is a **spring**, written with the force through it and the velocity across it. Current still plays the part of force, voltage of velocity, and $\frac{1}{2}Li^2$ matches the spring's $F^2/2k$. A stiff spring is a small inductance.

7. The LTI elements, side by side

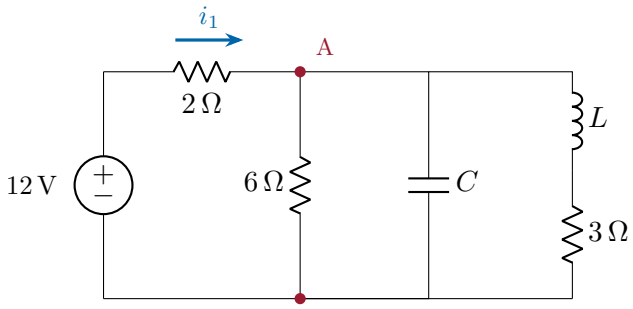
That is the whole cast for now. Three elements, one line each.

	LTI resistor	**LTI capacitor**	**LTI inductor**
characteristic lives in	v–i	v–q	i–φ
that characteristic	$v = Ri$	$q = Cv$	$\varphi = Li$
terminal equation	$v = Ri$	$i = C\,dv/dt$	$v = L\,di/dt$
integral form	—	$v = v_0 + \frac{1}{C}\int i\,d\tau$	$i = i_0 + \frac{1}{L}\int v\,d\tau$
unit	ohm, Ω	farad, F	henry, H
memory	none	the past of i	the past of v
at DC it is	itself	an *open* circuit	a *short* circuit
cannot jump	—	its voltage	its current
energy w	—	$\frac{1}{2}Cv^2$	$\frac{1}{2}Li^2$
power p	$Ri^2 \geq 0$	either sign	either sign
verdict	passive, *lossy*	passive, *lossless*	passive, *lossless*

Read the last two columns against each other and the duality is complete: swap $v \leftrightarrow i$, $q \leftrightarrow \varphi$, $C \leftrightarrow L$, and every line turns into the other.

8. A circuit with all three

The two DC rules in the table solve a circuit outright, provided nothing is changing. Here is Lecture 2's circuit with a capacitor and an inductor added, long after the switch was closed.



$C = 100\,\mu\text{F}$, $L = 50\,\text{mH}$.

Nothing has changed for a long time, so every derivative is zero.

Example 5.3 — (a) find every branch variable in the steady state

Step 1 — replace the two stores. Nothing is changing, so by the table the capacitor is an open circuit and the inductor a short circuit. What is left is Lecture 2's resistive circuit.
Step 2 — solve it. The 6 Ω and 3 Ω branches both run from node A to the bottom node, so

$$i_1 = \frac{12}{2 + (6 \parallel 3)} = \frac{12}{2+2} = 3\,\text{A}, \qquad v_A = 12 - 2(3) = 6\,\text{V}.$$

$$i_{6\Omega} = \frac{6}{6} = 1\,\text{A}, \quad i_L = i_{3\Omega} = \frac{6}{3} = 2\,\text{A}, \quad v_C = 6\,\text{V}, \quad v_L = 0, \quad i_C = 0,$$

$$w_C = \tfrac{1}{2}(10^{-4})(6)^2 = 1.8\,\text{mJ}, \qquad w_L = \tfrac{1}{2}(0.05)(2)^2 = 0.1\,\text{J}.$$

Example 5.3 — (b) where is the power going?

$$p_{\text{source}} = -12 \times 3 = -36\,\text{W (delivered)}, \qquad p_{2\Omega} = 2(3)^2 = 18\,\text{W},$$

$$p_{6\Omega} = \frac{6^2}{6} = 6\,\text{W}, \qquad p_{3\Omega} = 3(2)^2 = 12\,\text{W}.$$

The resistors dissipate $18 + 6 + 12 = 36\,\text{W}$, exactly what the source delivers. And the two stores? The capacitor carries no current, the inductor has no voltage across it, so

$$p_C = v_C\, i_C = 6 \times 0 = 0, \qquad p_L = v_L\, i_L = 0 \times 2 = 0.$$

They hold energy. They do not consume any.

The 1.8 mJ and the 0.1 J just sit there. Change something and they speak — which is what the next few lectures are about.

Summary

- An **inductor** fights changes in current, with a **back EMF** $v = L\,di/dt$. L is in **henries**, set by the turn count, the winding's shape and what is inside it; one henry is large.
- Integrate the voltage: $\varphi = \int v\,d\tau$ in **volt-seconds** (webers), exactly as integrating the current gave charge in amp-seconds. An inductor is a curve in the i–φ plane; the **LTI** one is $\varphi = Li$. Not a resistor — no f with $v(t) = f(i(t))$.
- Short circuit at DC; its **current is continuous**; it remembers.
- Starting the current means **work against the back EMF**, and the total $w = \frac{1}{2}Li^2$ is kept in the magnetic field and handed back in full: passive and *lossless*. Forcing the current to jump demands a voltage impulse — in a real switch, a spark.
- In a **DC steady state** the capacitor is open and the inductor short, so the circuit collapses to a resistive one and neither store absorbs power. Swap $v \leftrightarrow i$, $q \leftrightarrow \varphi$, $C \leftrightarrow L$: the two stores are one element told twice (Section 7).

Next lecture

Dependent sources — whose value is set by a voltage or current elsewhere in the circuit. They are how a transistor or an op-amp gets into a circuit diagram.